

Santhosh Narayanan Baburaman

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EDUCATION

M.S. Analytics, University of Southern California (GPA: 3.92/4.0) Jan 2025 – Dec 2026

Coursework: Predictive Analytics, Financial Analytics, Data Engineering, Decision Analysis

B.Tech. Computer Science and Engineering, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024

Publication: Co-authored an IEEE conference paper on predictive analytics for transportation, modeling travel time, fare, and traffic speed across 1M+ NYC taxi trip records

EXPERIENCE

Data Analyst Intern | Philadelphia Department of Aviation Jun 2026 – Aug 2026

- Designed the department's first structured reporting and data-verification layer across 2 revenue streams covering ~30% of airport operating revenue, replacing ~60 hours/month of manual review and presenting results to senior management each cycle
- Engineered a Python → Parquet → Alteryx → Power BI pipeline over 800K+ rows of billing data spanning 4 fiscal years, replacing manual Excel review (~40 hrs/month) and surfacing anomalies the prior process missed
- Rebuilt ground transportation reporting on PropWorks SQL → Amazon Redshift into Power BI, correcting aggregation-order and duplicate-rate defects that had misstated fiscal-year totals and replacing a SQL backend impractical to audit directly (~20 hrs/month saved)
- Integrated both data models to compute per-mode passenger capture rates (ground transportation vs. passenger arrival/departure volumes) and a seasonal Z-score anomaly detector with peer-deviation and Benford's Law checks, enabling curbside staffing and audit-style error correction

Founder & President | ISE Analytics Student Association (IASA), USC Feb 2025 – Present

- Founded USC's first analytics student organization, hosting 12+ events for 500+ students and 20+ professionals

PROJECTS

Consulting Firm Data Warehouse (dbt, Snowflake, SQL, Python)

- Built a multi-layer ELT pipeline in dbt on Snowflake, transforming 17 normalized source tables into 3 subject-area datamarts for a 4-person team
- Modeled 7 conformed dimensions including an SCD Type 2 dimension preserving salary and title history, plus a generated date spine for consistent time aggregation
- Wrote 39 automated data-quality tests and reconciled design docs against the production schema, documenting 13 discrepancies for the team

Financial Analytics & Power BI Dashboard (Python, PostgreSQL, Power BI) 

- Built an ETL pipeline extracting 10-K/10-Q data via the SEC EDGAR API into a PostgreSQL star schema, delivering a Power BI dashboard visualizing multi-company revenue, profit, and asset trends (2010–2025)

Automated Sales Analytics Copilot (Python, PostgreSQL, Claude API, Streamlit) 

- Designed a PostgreSQL star schema with 5 KPI views tracking revenue, margin, and discount impact across categories and regions on a 10K-order, \$2.3M dataset
- Automated anomaly detection and forecasting across 4 years of data, surfacing that Furniture margins collapse to -22% under high discounts, with nightly Slack alerts
- Built a natural language chat UI letting business users query KPIs, run what-if scenarios, and generate ad-hoc reports without writing SQL

Sales Performance & Profitability Dashboard (Tableau) 

- Built an interactive Tableau dashboard analyzing \$12M+ in sales, identifying negative margins and losses from high discounting

NYC Taxi Demand & Fare Analysis (Python, Time Series, Feature Engineering) 

- Analyzed 1M+ NYC taxi trip records to model travel time, fare, and traffic speed with engineered temporal features (see IEEE publication above)

SKILLS

- **Data & Analytics:** Python, SQL, R, Pandas, ETL/ELT, Data Modeling, Data Quality Testing, KPI Design
- **BI & Platforms:** Power BI (DAX, Power Query), Tableau, Streamlit, PostgreSQL, Snowflake, Redshift, dbt, Alteryx
- **Methods & Cloud:** Predictive Modeling, Time Series, Anomaly Detection, GCP, LLM APIs, Git